

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

KIDS-ELEM2 — Full working manuscript

Age guide: Grades 3–5/6

Review prototype · not child-validated · not publication-ready

S01 — U01 · OBSERVE — The screen is not the whole system



figures/FIG-ELEM2-U01-S01.svg

Mira opens a familiar app and asks the crew to name what is **visible**: icons, text, sound, vibration. Bolt points past the glass: the screen is one surface of a larger system. On a shared map, label layers you can honestly name — people and intent, input hardware, operating system, application, optional network path, output, your perception. Circle what you truly observed. Put a question mark on what you only infer. A map is a model: useful, incomplete, and revisable. If two classmates mark different question marks, that is good science, not a fight.

Try it: Annotate a visible→hidden systems map; mark observe vs infer.

S02 — U01 · PREDICT — Local or farther away?



diagram

Ping asks: for **this** action, does anything need to leave the device? Predict in writing before you touch anything. Local-only actions can finish with power, CPU work, memory, and storage on the device. Networked actions may need packets on a path. Write three boxes: prediction → planned observation → what would count as evidence. Do not treat “it felt instant” as proof that no network was involved.

Try it: Write a local-vs-remote prediction for one familiar action.

S03 — U01 · TEST — Airplane mode as an honest tool



diagram

With an adult, test one safe action with radio off (airplane mode) if the device allows it. What still works? What fails, hangs, or stays blank? Record the result next to your prediction. If you cannot test safely today, keep the prediction and write why the test waited. Evidence beats slogans.

Try it: One supervised radio-off observation, or a written “test waited” note.

S04 — U01 · MEASURE — Felt delay vs measured wait



figures/FIG-ELEM2-U01-S05.svg

Latency means waiting time somewhere on a path. Felt delay is what your body notices; measured delay is what a classroom timer roughly shows. On a remote refresh (adult-approved), mark start, first visible change, and “done enough.” Write units (seconds). Separate felt delay from guessed cause — busy CPU, full memory, slow storage, weak Wi-Fi, or a far server can feel similar until you gather evidence. Rough numbers beat confident myths.

Try it: Three rough stopwatch marks on one approved action.

S05 — U01 · EXPLAIN — CPU, memory, storage are jobs



figures/FIG-ELEM2-U01-S02.svg

Bolt deals three role cards. **CPU** runs instructions right now. **Memory** (working space) holds state the program is using this moment. **Storage** (keep-box) keeps durable data for later. Sort classroom examples: instructions being followed, today's draft still open, yesterday's saved file. Say the roles aloud. A bigger advertisement number does not tell the whole story — ask what was actually measured. If a label sounds like an ad for “brain speed,” rewrite it as a job the machine can do.

Try it: Sort role cards for CPU / memory / storage with real examples.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S06 — U01 · BUILD — Paper systems protocol



diagram

Build a one-page protocol for tracing one experience across layers: numbered steps, a failure branch (“if no change, check power / ask a trusted adult”), and a short teach script for someone younger. Include one blank for a repair you will make after testing.

Try it: Protocol draft + younger teach script stub.

S07 — U01 · SECURE — Who can see this path?



Shield asks two calm questions about your mapped action: What security control protects the device or account (lock screen, update, permission prompt)? What privacy choice changes who can see information? List one of each. We learn protection tools — we do not practice breaking into systems.

Try it: One security control + one privacy choice tied to your map.

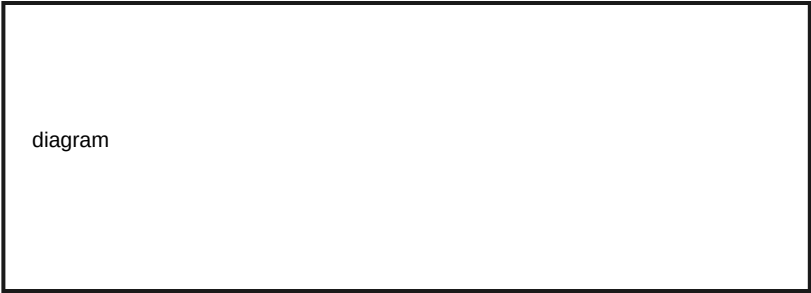
S08 — U01 · REFLECT — Spiral: same idea, more tools



What stays the same from earlier “touch → change” learning to this systems map?
Contingent action and response. What vocabulary was earned later: input, output, event, packet, latency, privacy, security? Write one sentence you would teach a younger learner and one sentence only this class needs.

Try it: Spiral reflection note (younger / this class).

S09 — U01 · TEACH — Map teach-back



diagram

Teach a partner your systems map. Name what you observed, what you inferred, and what you still do not know. Invite one question you cannot answer yet and write it down for next time. Do not invent evidence to sound finished.

Try it: Partner teach-back with observed / inferred / unknown.

S10 — U02 · OBSERVE — Energy makes work possible



figures/FIG-ELEM2-U02-S01.svg

Bolt shows a battery icon and a warm laptop edge (look, don't press hard). Devices need energy to run instructions, light a display, talk on radio, and keep storage alive.

Warmth can be a clue that work is happening — not a magic mood. Observe: charge level (if shown), fans or silence, and whether an adult says the device feels only warm or too hot.

Try it: Observation sheet: energy clue + warmth clue + adult note.

S11 — U02 · PREDICT — What costs energy?



diagram

Predict which classroom actions spend more energy: bright screen vs dim; local note app vs long video stream; radio off vs radio on. Rank three actions from “probably less” to “probably more.” Write why in one sentence each — still a prediction, not a proof.

Try it: Rank three actions by predicted energy cost.

S12 — U02 · TEST — Dim vs bright, short trial



diagram

With adult permission, run two short, stoppable trials on the same device: dim display for one minute of a local task, then bright for one minute of the same local task. Note battery percentage only if the OS shows it honestly — do not force settings you don't understand. Record what changed and what stayed the same.

Try it: Two one-minute supervised trials; note battery if visible.

S13 — U02 · MEASURE — Heat as a stop rule

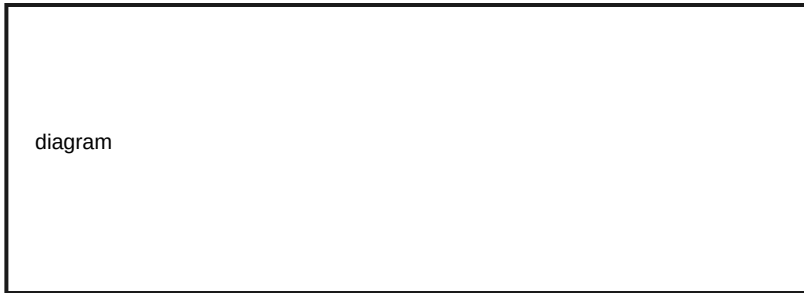


figures/FIG-ELEM2-U02-S04.svg

Measure with senses and adult judgment, not with homemade “temperature experiments” on gadgets. If a device is uncomfortably hot, stop using it and tell an adult. Write a classroom stop rule in your own words: warm may mean work; hot means pause. Adults handle chargers and damaged cables.

Try it: Write a thermal stop rule; role-play asking an adult.

S14 — U02 · EXPLAIN — Constraints shape design



Power budgets and heat limits shape what engineers can build: how bright, how fast, how long a battery lasts, how thick a case can be. Interconnects (paths that carry signals and power between parts) also constrain design — parts must connect safely. We describe constraints, not brand winners.

Try it: List two constraints (power/heat/interconnect) that could change a design.

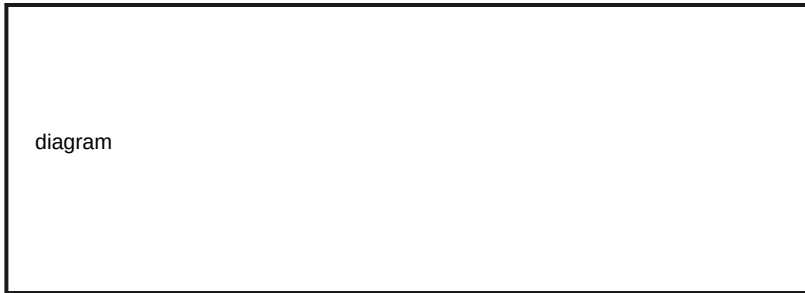
S15 — U02 · BUILD — Constraint cards



Build three paper cards: Power, Heat, Interconnect. On each card, write one observation from this unit and one design question (“What would we change if...?”). Keep questions open — engineers revise.

Try it: Three constraint cards with observation + question.

S16 — U02 · SECURE — Safe handling habits



Shield's checklist: adult handles chargers; don't use damaged cables; don't block vents on purpose; stop if hot; no opening devices. Security of the machine starts with not creating hazards while you learn.

Try it: Copy the safe-handling checklist into your notebook.

S17 — U02 · REFLECT — Limits are information



diagram

How does knowing about power and heat change how you explain a “slow” device? Maybe it is throttling work to stay cool — an inference you should mark with a ?. Write one revised explanation that includes a constraint.

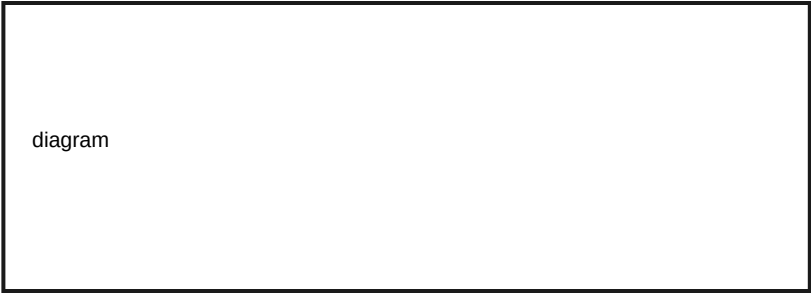
Try it: Rewrite one “slow” story with a possible constraint + ?.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S18 — U02 · TEACH — Constraint teach-back



diagram

Teach a younger learner: devices need energy; warmth can mean work; hot means ask an adult. Use fewer new words. Then teach a peer one interconnect idea: parts need paths for power and signals.

Try it: Two teach scripts (younger / peer).

S19 — U03 · OBSERVE — A tap becomes an event



figures/FIG-ELEM2-U03-S01.svg

Step narrates: when you tap, input hardware reports activity. The operating system notices and delivers an **event** toward the app that should respond. Inside the app, a handler can run — instructions that update **state** (the program's current facts) and ask for new output. Draw one chain on a single page: tap → OS event → app handler → updated state → screen or sound. This is a classroom model — useful, incomplete, revisable.

Try it: Draw the event-handling sequence on one page.

S20 — U03 · PREDICT — Which handler fires?



diagram

Predict: if you tap Save versus Undo in a writing app, which state changes? Predict outputs you expect to see. Write “if this event, then this handler idea” in plain words — you are designing a tiny mental program, not claiming you know the vendor’s code.

Try it: Two if-event → then-handler predictions.

S21 — U03 · TEST — UI as conversation with a program



figures/FIG-ELEM2-U03-S03.svg

A user interface (UI) is how people send events and read outputs. Test one supervised app: trigger two different events and record state changes you can see (text appeared, button grayed out, file name changed). If nothing visible changes, write that honestly — invisible work may still be happening, but you must not invent it.

Try it: Two events → visible state notes.

S22 — U03 · MEASURE — Order and timing of steps

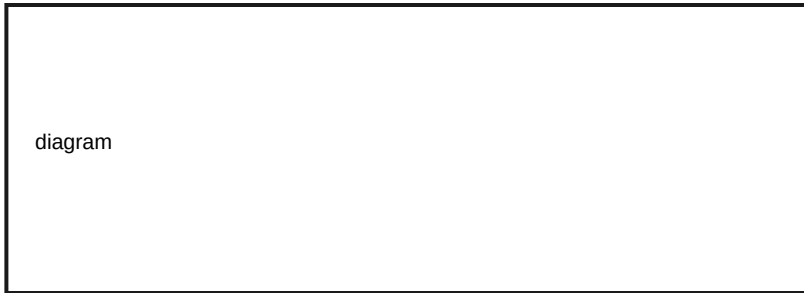


diagram

Instructions have order. Measure roughly: how many distinct visible steps between your event and a finished-looking result? Count frames of change if you can (1...2...3). Programming is giving ordered instructions a machine can follow; bugs often mean the order or the conditions are wrong.

Try it: Count visible steps from event to “done enough.”

S23 — U03 · EXPLAIN — Files, data, and durable state



Some state lives only in memory while the app is open. **Files** and durable storage keep data for later. When you save, you ask the system to write lasting data. When you open, you read it back. Trace: event Save → handler → storage write → later event Open → handler → memory holds the data again for editing.

Try it: Trace save/open as events that touch storage.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S24 — U03 · BUILD — Paper event app

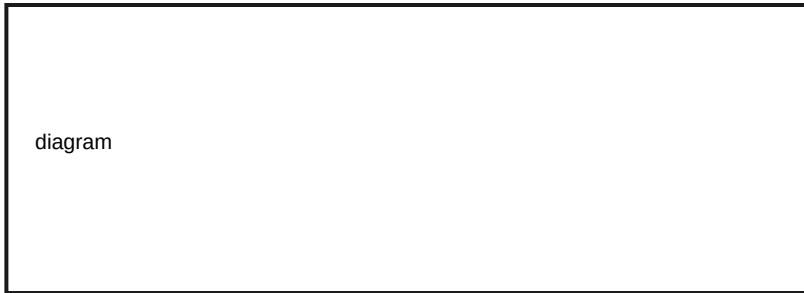


diagram

Build a paper mini-app: three events (Tap A / Tap B / Reset), a state box (score or mode), and handlers written as short sentences. Act it out with a partner as the “runtime.” Repair one unclear handler after the first run.

Try it: Paper app + one repair after rehearsal.

S25 — U03 · SECURE — Permissions are events too



diagram

Shield points at permission prompts: an app may request camera, mic, or location. That prompt is a security and privacy event — your choice (with an adult) can allow, deny, or ask later. List one permission you would want an adult to decide.

Try it: Name one permission decision for an adult.

S26 — U03 · REFLECT — Programming as revise-able plans



diagram

How is fixing a paper handler like debugging? You change instructions, retest, and keep evidence of the repair. Write one sentence: “My first handler failed because...; I improved it by...”

Try it: Failure → repair sentence.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S27 — U03 · TEACH — Event chain teach-back



diagram

Teach someone the chain without stack jargon overload: something happens → the system notices → instructions run → something updates → you see or hear a result. Add one honest limit: we do not see every step inside.

Try it: Teach-back with one named limit.

S28 — U04 · OBSERVE — Messages can be split



figures/FIG-ELEM2-U04-S01.svg

Ping cuts a paper letter into addressed pieces called ****packets**** in our classroom model: little message pieces with enough addressing to travel and reassemble. Observe a relay game: sender → hop → hop → receiver. Some pieces may arrive out of order; some may wait. Reliability is not guaranteed just because sending felt easy.

Try it: Packet relay with addressed paper pieces.

S29 — U04 · PREDICT — Path and failure



diagram

Predict two failure modes: a piece goes missing; a hop is slow. What would the receiver notice? What could the sender try next (resend, wait, ask an adult)? Write predictions before the relay.

Try it: Missing-piece and slow-hop predictions.

S30 — U04 · TEST — Wi-Fi is not the whole Internet



figures/FIG-ELEM2-U04-S03.svg

Wi-Fi is often a local radio hop between your device and a nearby access point. The **Internet** is a huge collection of networks that can carry packets farther. Cellular is another kind of radio access road. Test ideas with labels, not myths: “Connected to Wi-Fi” does not mean “I am the Internet.” With an adult, name which hop you can actually identify today.

Try it: Label Wi-Fi hop vs farther Internet path on a diagram.

S31 — U04 · MEASURE — Latency and reliability notes



diagram

Time a classroom packet relay: start when first piece leaves; end when the message is interpretable. Count pieces lost. Record latency (wait) and reliability (how complete). Felt lag on a real device may mix radio conditions, routing distance, and busy servers — mark causes with ? until tested.

Try it: Relay timer + lost-piece count.

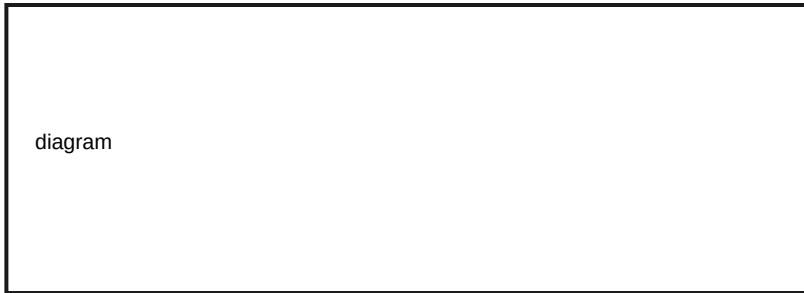
S32 — U04 · EXPLAIN — Represent → transmit → interpret



Data is represented (symbols/bits in our story), transmitted (moved), routed/forwarded (handed along paths), received, and interpreted (turned back into meaning for a person or program). If interpretation fails, you may see nonsense or an error — that is information too.

Try it: Five-box flow: represent / transmit / route / receive / interpret.

S33 — U04 · BUILD — Classroom routing map



Build a floor map with three hops and two possible routes. Send one packet each way. Note which route felt longer. Optional: mark a “congested” hop that must wait.

Try it: Two-route map with timing notes.

S34 — U04 · SECURE — Messages and strangers



Shield: do not send personal information in classroom message games. Real networks can expose messages to more places than you see. We may say encrypted paths try to make messages harder for strangers to read in transit — without practicing attacks. Adult mediation for any real messaging.

Try it: List two things that should not ride on a practice packet.

S35 — U04 · REFLECT — Sensors and far paths



diagram

Some devices use **sensors** (light, motion, touch) to create local events; some sensor readings may later be sent as packets. Reflect: which of today's examples stayed local, and which needed a route? Mark unknowns.

Try it: Local vs routed sensor/message table.

S36 — U04 · TEACH — Packet story for a peer



Teach a peer using the letter metaphor, then retire the metaphor: real packets are not paper, but addressing, hops, delay, and loss still matter. End with Wi-Fi ≠ Internet.

Try it: Teach-back ending in the Wi-Fi misconception fix.

S37 — U05 · OBSERVE — Patterns from examples



figures/FIG-ELEM2-U05-S01.svg

Mira sorts example cards (shapes, words, or animal pictures). A **model** in this unit means a tool that found patterns in data and can output a **prediction** or generated suggestion. Observe carefully: which features did your sorting rule use (color, size, first letter)? Write the rule in one sentence. With few examples, guesses are shaky; with biased examples, guesses can be unfair. People choose data and checks. Sensors can create data too — a light sensor reading is still just a measurement until someone interprets it.

Try it: Sort examples; write the sorting rule; note what a classmate might “learn.”

S38 — U05 · PREDICT — When will it fail?



diagram

Predict a failure: show the “model” (a classmate’s sorting rule) a case outside the training examples. What wrong label might appear? Write the prediction before the test.

Try it: Out-of-example failure prediction.

S39 — U05 · TEST — Inference is not understanding



figures/FIG-ELEM2-U05-S04.svg

****Inference**** here means running a model to get an output now. Test your classmate-rule model on five new cards. Score hits and misses. Important honesty: models do not “understand” like people — they match patterns from data. Fluent text from a generative tool can still be wrong. No open-ended child–AI chat is required for this lab; paper models are enough.

Try it: Five-card test with hit/miss tally.

S40 — U05 · MEASURE — Error rate, rough and honest



diagram

Measure: misses $\div$ total trials. Write the fraction and a plain sentence (“About 2 of 5 wrong on this tiny set”). Tiny sets do not prove a product is good or bad worldwide — they teach the habit of measuring.

Try it: Compute a tiny-set error rate with units.

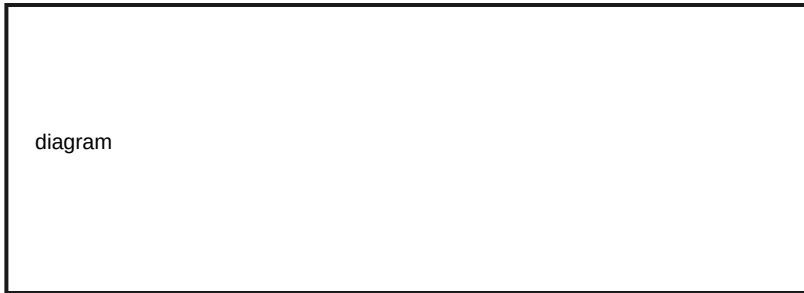
S41 — U05 · EXPLAIN — Pipeline and responsibility



Explain the pipeline: data → model → prediction/generation → human check → decision. Sensors may feed data; storage may keep datasets; networks may move them. At each arrow, ask who is accountable. Shield adds: if an output affects a person, a person must be able to question it.

Try it: Draw the pipeline; star the human-check step.

S42 — U05 · BUILD — Check protocol



Build a three-question check protocol for any model output: (1) What evidence supports this? (2) What could be missing from the data? (3) Who should confirm before we trust it? Keep it on a card for Unit 7's portfolio.

Try it: Three-question check card.

S43 — U05 · SECURE — Data about people



diagram

If data is about people, privacy and fairness rules tighten. Do not collect classmate personal data for a “fun dataset.” Use public classroom-safe examples only. Equity means noticing who is left out of the examples.

Try it: Rewrite a bad data idea into a privacy-safe version.

S44 — U05 · REFLECT — Careers precursor

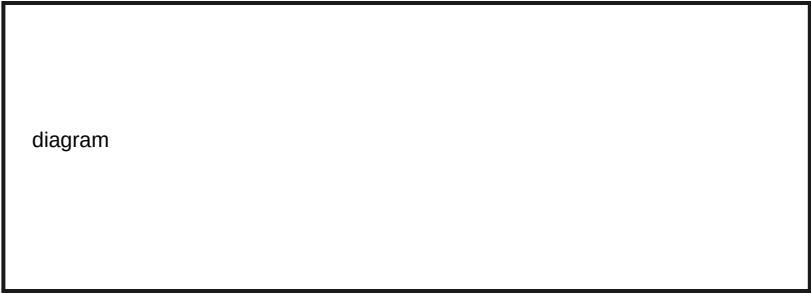


diagram

People who work with data and models test, measure, document limits, and teach others what can go wrong. That is a career precursor — curiosity plus honesty — not a job title quiz. Write one skill you practiced today that belongs in a young-engineer portfolio.

Try it: One portfolio skill sentence.

S45 — U05 · TEACH — Limits teach-back



diagram

Teach: models predict from patterns; people check; models don't know like persons. Offer one example of a wrong but fluent answer you invented on paper to practice catching it.

Try it: Teach-back + one deliberate wrong fluent example to catch.

S46 — U06 · OBSERVE — Three related ideas



figures/FIG-ELEM2-U06-S01.svg

Shield separates ideas that often get mashed together. **Security:** controls that protect systems and accounts — lock screens, updates, permission prompts. **Privacy:** choices about who can see or keep information — share settings, what you post, what an app stores. **Equity:** who can access, learn, and benefit — and who is excluded by missing devices, offline time, or inaccessible design. Observe classroom examples of each without ranking anyone's home devices. Write one concrete example per column from school life only.

Try it: Three-column notes: security / privacy / equity examples (school-safe).

S47 — U06 · PREDICT — Control vs choice vs access



diagram

Predict how a lock screen (security), a share setting (privacy), and a missing school laptop cart (equity) change what a student can do tomorrow. Write one sentence each.

Try it: Three predictions tied to the triad.

S48 — U06 · TEST — Permission and pause



diagram

Test a pause habit: before installing or signing into anything, stop and ask an adult.

Role-play two prompts: “Allow tracking?” and “Share location?” Practice saying “Ask a trusted adult.” This is a behavioral test of the pause — not a real account creation.

Try it: Two pause role-plays.

S49 — U06 · MEASURE — Access gaps without blame



diagram

Measure carefully: count classroom resources that are shared (loaner devices, library hours, offline packets). Equity work asks what people need to join — not who “failed” to buy gear. Write one measurable access support (hours, seats, offline option) without naming families.

Try it: One access-support measure, anonymized.

S50 — U06 · EXPLAIN — Chip-to-cloud duties (conceptually)



diagram

Duties stack: device locks, app permissions, account rules, network protections, and organizational policies. You do not need every layer's jargon — you need the idea that protection is layered, and privacy choices appear at multiple layers. Defense-in-depth is a calm concept: more than one control.

Try it: List three layers where a privacy or security choice can appear.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S51 — U06 · BUILD — Fair classroom tech charter



diagram

Build a short charter: stop anytime; ask adult for accounts; no sharing passwords; include offline ways to join; no ranking home devices; repair mistakes without humiliation. Sign with initials if you want — optional.

Try it: Classroom fair-tech charter draft.

S52 — U06 · SECURE — Deletion and copies



figures/FIG-ELEM2-U06-S05.svg

Deleting a file on one device may not remove every copy on backups, shared drives, or someone else’s download. Say what you can verify (“I deleted the classroom copy I control”) and what you only infer. Ask an adult about real deletion needs.

Try it: Write verify vs infer for a deletion story.

S53 — U06 · REFLECT — Who benefits?



diagram

Reflect on a tool you use: who benefits most, who might be left out, and what would make joining fairer? Keep tone respectful — systems can improve.

Try it: Benefits / left out / fairer idea.

S54 — U06 · TEACH — Triad teach-back



diagram

Teach a peer the triad with one example each. End with: we protect people; we do not practice harm.

Try it: Triad teach-back + non-harm closer.

S55 — U07 · OBSERVE — What evidence do you already have?



figures/FIG-ELEM2-U07-S01.svg

Gather artifacts from Units 1–6: systems maps, timers, constraint cards, paper apps, packet notes, model checks, fair-tech charter. Observe what is complete vs missing. A young-engineer ****portfolio precursor**** shows process — protocols, measures, repairs, teach scripts — not a fake certificate and not a ranked scoreboard. Put each artifact in an EMIT pile: useful for Explain, Measure, Improve, or Teach. If a pile is empty, that is your next build target.

Try it: Inventory checklist + EMIT pile sort of prior artifacts.

S56 — U07 · PREDICT — Where will the teach-back fail?



diagram

Predict one step a younger learner will find unclear in your best protocol. Plan a clarification before you teach.

Try it: Predicted confusion + planned fix.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S57 — U07 · TEST — Rehearse once



diagram

Test your teach script once with a partner. Mark where they stumble. Do not blame them — repair the script.

Try it: One rehearsal + stumble marks.

S58 — U07 · MEASURE — One wait, one error, one repair



diagram

Measure three portfolio numbers if you can: one latency (seconds), one error rate from the model lab (fraction), one repair count (how many script fixes). Label units. If a measure is missing, write why.

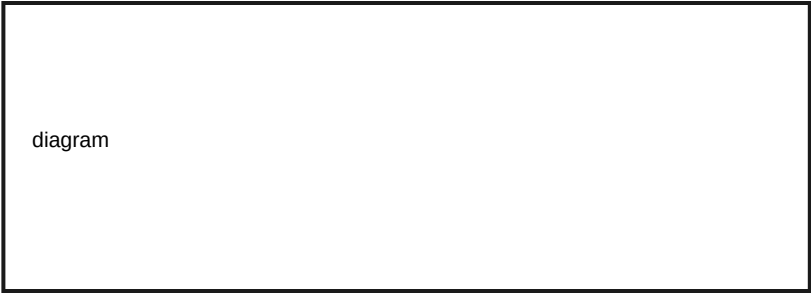
Try it: Three labeled measures or explicit gaps.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S59 — U07 · EXPLAIN — EMIT loop



diagram

****EMIT:**** Explain one path or system · Measure one quantity · Improve one step · Teach someone else. Explain your chosen path with observed vs inferred columns. Admit unknowns.

Try it: EMIT explanation page with ? marks.

**KIDS FULL WORKING MANUSCRIPT
NOT CHILD-VALIDATED
NOT PUBLICATION-READY**

S60 — U07 · BUILD — Work packet



figures/FIG-ELEM2-U07-S08.svg

Assemble a work packet: (1) systems or event protocol, (2) one measurement sheet with units, (3) model check card, (4) fair-tech charter snippet, (5) teach script for a chosen audience, (6) dated repair note describing what changed after rehearsal. Optional cover: your name or initials, date, and one sentence on what you still do not know. This packet is your young-engineer portfolio precursor — evidence of practice across systems, hardware limits, instructions, messages, data checks, and safety/equity.

Try it: Six-part work packet assembly (+ optional cover unknowns sentence).

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S61 — U07 · SECURE — Share safely



diagram

Before sharing a packet photo or digital copy, Shield checks: no passwords, no personal addresses, no classmates' private data, adult OK for any upload. Prefer paper gallery walks.

Try it: Pre-share safety checklist initialed by adult if sharing beyond class.

S62 — U07 · REFLECT — Careers and next questions



diagram

Reflect: which role felt closest today — mapping systems, hardware constraints, instructions, networks, data checks, safety/equity, or teaching? Write two next questions for a future lab. Careers grow from repeated honest practice.

Try it: Role reflection + two next questions.

S63 — U07 · TEACH — Capstone teach-back



Teach your EMIT story to someone else. Use the systems map or event chain as a shared figure. Name what you observed, what you inferred, and what you still do not know. Invite questions you cannot answer yet; write them down for a future lab. Do not invent evidence to sound finished. If energy drops, stop — an easy stop is part of good teaching. Careers in technology grow from repeated honest practice: explain, measure, improve, teach.

Try it: Final teach-back with unknown list and optional stop.